Which Proteins are Synthesised?

Exam code: 0610 & 0970

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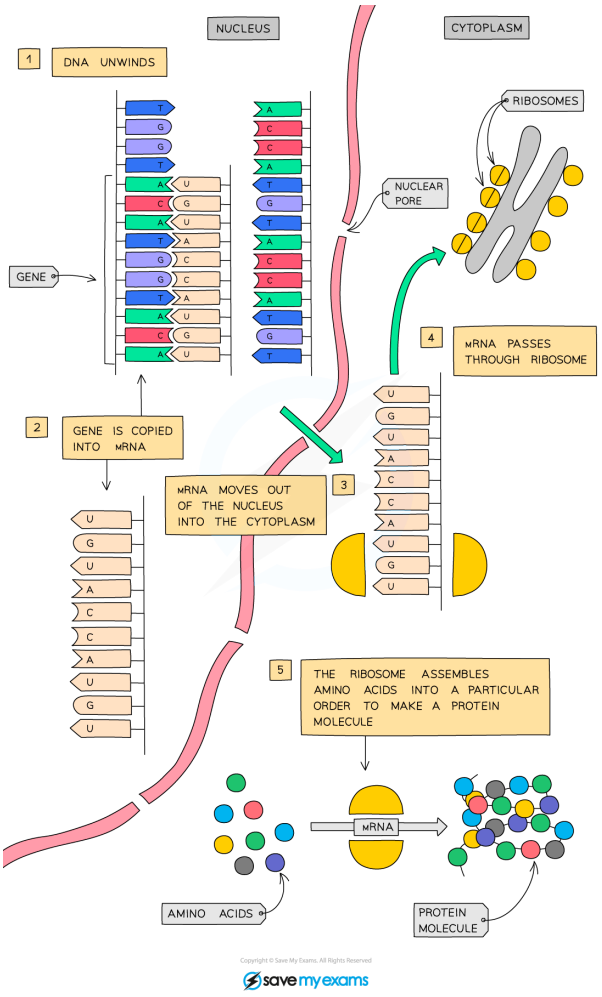
Control of Gene Expression: Extended



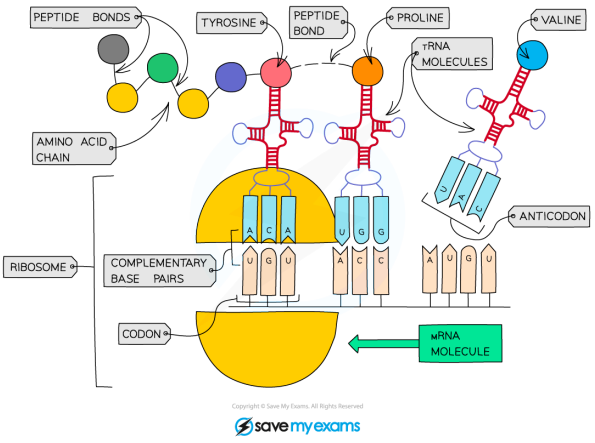
Protein synthesis

- **Expression** of a gene means whether that gene is transcribed and translated in a particular cell or not
- Most genes are not expressed in a particular cell
  - They are 'switched off'

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

function are expressed ('switched on')



The triplet code of DNA (carried by mRNA) is read by the ribosome and amino acids are attached together in a specific sequence to form the protein

- In this way, DNA **controls cell function** by **controlling the production of proteins**
- The proteins may be **enzymes, antibodies,** or **receptors** for **neurotransmitters**
- Although all body cells in an organism contain the same genes, **many genes in a particular cell are not expressed** because **the cell only makes the specific proteins it needs**

Haploid and Diploid Cells

- All humans have **23 different chromosomes** in each cell
- In most body cells, not including the gametes (sex cells), we have 2 copies of each chromosome, leading to a total of **46 chromosomes**
- Nuclei with two sets of chromosomes are known as **diploid nuclei**
- The **gametes** (egg and sperm cells) only have one copy of each chromosome, meaning they have a **total of 23 chromosomes** in each cell
- Nuclei with one set of unpaired chromosomes are known as **haploid nuclei**

Examiner Tips and Tricks

An easy way to remember the difference between haploid and diploid is to remember:

Haploid = **Ha**lf the normal number of chromosomes

It's worth noting that the human diploid chromosome number is 46. In an exam, you may be given information about a

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